Chapter 16
Conceptual Database Design for the Relational Model

Objectives

- The purpose of a design methodology.
- The 3 main phases of database design:
  - conceptual, logical, and physical design.
- Document the process of conceptual database design

Database Design

- In Chapter 10 we described the main stages of the database system development lifecycle, one of which is database design.
- Database design is made up of 3 phases:
  - Step 1: Conceptual
  - Step 2: Logical
  - Step 3: Physical database design
- Conceptual database design is to build the conceptual representation of the database, which includes identification of the important entities, relationships, and attributes.

What is a Design Methodology?

- Design methodology is a structured approach that uses procedures, techniques, tools, and documentation aids to support and facilitate the process of design.

Conceptual Database Design Methodology

- Step 1: Build Conceptual Data Model
- The objective is to build a conceptual data model of the data requirements of the enterprise.
- The conceptual data model is supported by documentation, including ER diagram and data dictionaries.

Step 1: Build Conceptual Data Model

Activities involved:

- Step 1.1 Identify entity types
- Step 1.2 Identify relationship types
- Step 1.3 Identify and associate attributes with entity or relationship types
- Step 1.4 Determine attribute domains
- Step 1.5 Determine candidate, primary, and alternate key attributes
- Step 1.6 Draw ER diagram. Consider the use of enhanced modeling concepts (optional).

Step 1: Build Conceptual Data Model (cont'd)

Activities involved:

- Step 1.7 Check model for redundancy and remove any that exists
- Step 1.8 Validate conceptual data model against user transactions
- Step 1.9 Review conceptual data model with user

Step 1.1 Identify entity types

- One method of identifying entities is to examine the users' requirements specification.
- As entity types are identified, assign them meaningful names.
- Record the names and descriptions of entities in a data dictionary.

Entity name	*Description*	*Aliases*	*Occurrence*
Staff	General term describing all staff employed by *DreamHome*.	Employee	Each member of staff works at one particular branch.
PropertyForRent	General term describing all property for rent.	Property	Each property has a single owner and is available at one specific branch, where the property is managed by one member of staff. A property is viewed by many clients and rented by a single client, at any one time.

Figure 16.1 Extract from the data dictionary for the StaffClient user views of DreamHome showing a description of entities.

Step 1.2 Identify relationship types

- Determine the multiplicity constraints of relationship types.
- Document relationship types and the multiplicity constraints in the data dictionary.

Entity name	*Multiplicity*	*Relationship*	*Multiplicity*	*Entity name*
Staff	0..1	*Manages*	0..100	**PropertyForRent**
	0..1	*Supervises*	0..10	**Staff**
PropertyForRent	1..1	*AssociatedWith*	0..*	**Lease**

Figure 16.3 Extract from the data dictionary for the StaffClient user views of *DreamHome*, showing a description of relationships.

Step 1.3 Identify and associate attributes with entity or relationship types

- Identify simple/composite attributes
- Identify single/multi-valued attributes
- Identify derived attributes
- Document attributes in the data dictionary

Primary key? Candidate key? Derived attribute?, etc…..

Entity name	*Attributes*	*Description*	*Data Type & Length*	*Nulls*	*Multi-valued*
Staff	**staffNo**	Uniquely identifies a member of staff	5 variable characters	No	No
	name				
	fName	First name of staff	15 variab e characters	No	No
	lName	Last name of staff	15 variab e characters	No	No
	position	Job title of member of staff	10 variab e characters	No	No
	sex	Gender of member of staff	1 character (M or F)	Yes	No
	DOB	Date of birth of member of staff	Date	Yes	No
PropertyForRent	**propertyNo**	Uniquely identifies a property for rent	5 variable characters	No	No

Figure 16.4 Extract from the data dictionary for the StaffClient user views of *DreamHome* showing a description of attributes.

Step 1.4 Determine attribute domains

- Document attributes domains in the data dictionary.
- Update the data dictionary entries for attributes to record their domain in place of the data type and length information.

Step 1.5 Determine candidate, primary, and alternate key attributes

- In the process of identifying primary keys, note whether an entity is strong or weak.
- If we are able to assign a primary key to an entity, it is referred to as being strong.
- Otherwise, the entity is referred to as being weak.
- The primary key of a weak entity can be identified only when we map the weak entity and its relationship with its owner entity to a relation through the placement of a foreign key in that relation (to be described in Step 2.1 of Chapter 17)
- Document primary and alternate keys in the data dictionary.

Step 1.6 Draw ER diagram

- In this step, we have the option to continue the development of the ER model using the enhanced modeling concepts, namely specialization / generalization.

